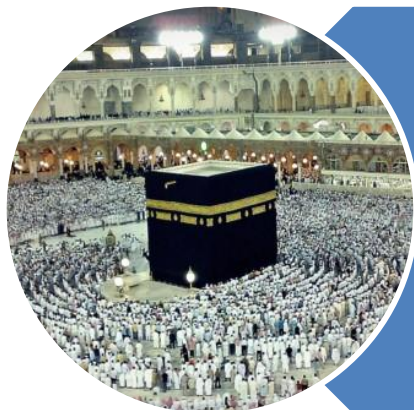


رَبِّ اشْرَحْ لِي صَدْرِي وَيَسِّرْ لِي أَمْرِي وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي



O my Lord! Expand for me my chest [with assurance] and ease for me my task and untie the knot from my tongue that they may understand my speech. **(Quran 20 : 25-28)**

Logarithmic Sorting

Merge Sort

C

P

C

S

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3

Merge Sort - **Algorithm**

- If the “list” is of length 0 or 1, then it is already sorted!
- Otherwise:
 1. Divide the unsorted list into two sub-lists of about half the size
 - So if your list has n elements, you will divide that list into two sub-lists, each having approximately $n/2$ elements:
 2. Recursively sort each sub-list by calling recursively calling Merge Sort on the two smaller lists
 3. Merge the two sub-lists back into one sorted list

C

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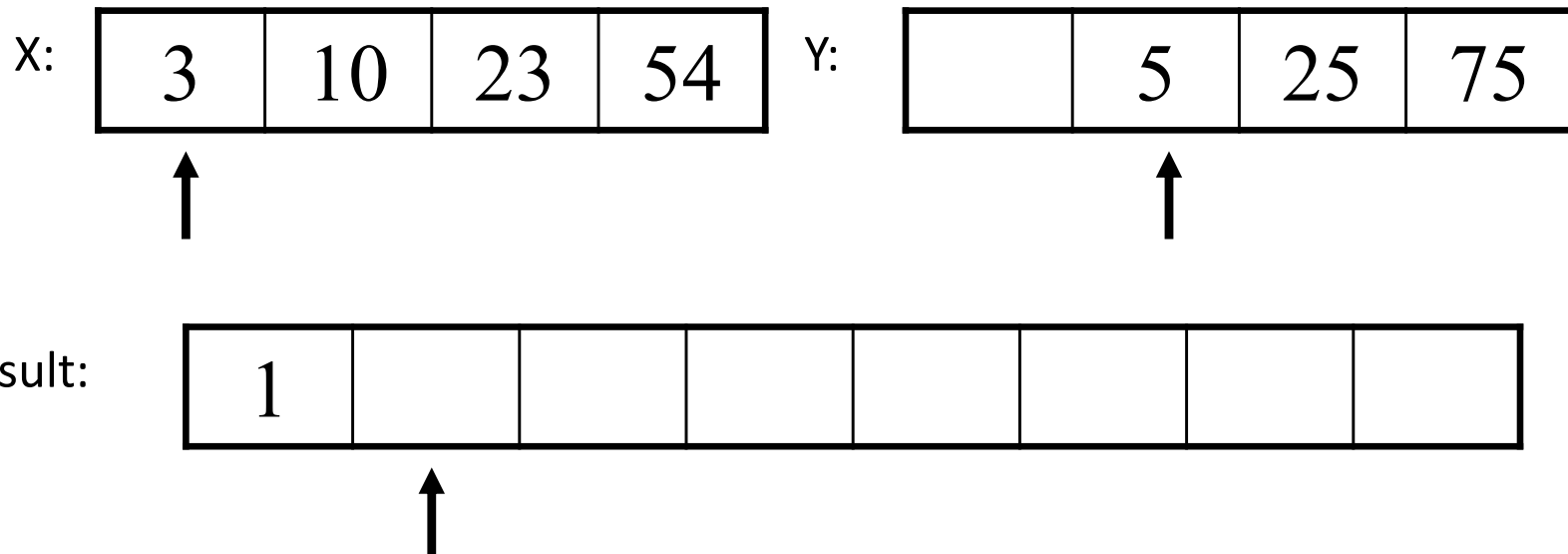
Merge Sort - **Algorithm**

```
method MergeSort(array) {
```

1. MergeSort(left half of array)
2. MergeSort(right half of Array)
3. Merge

```
}
```

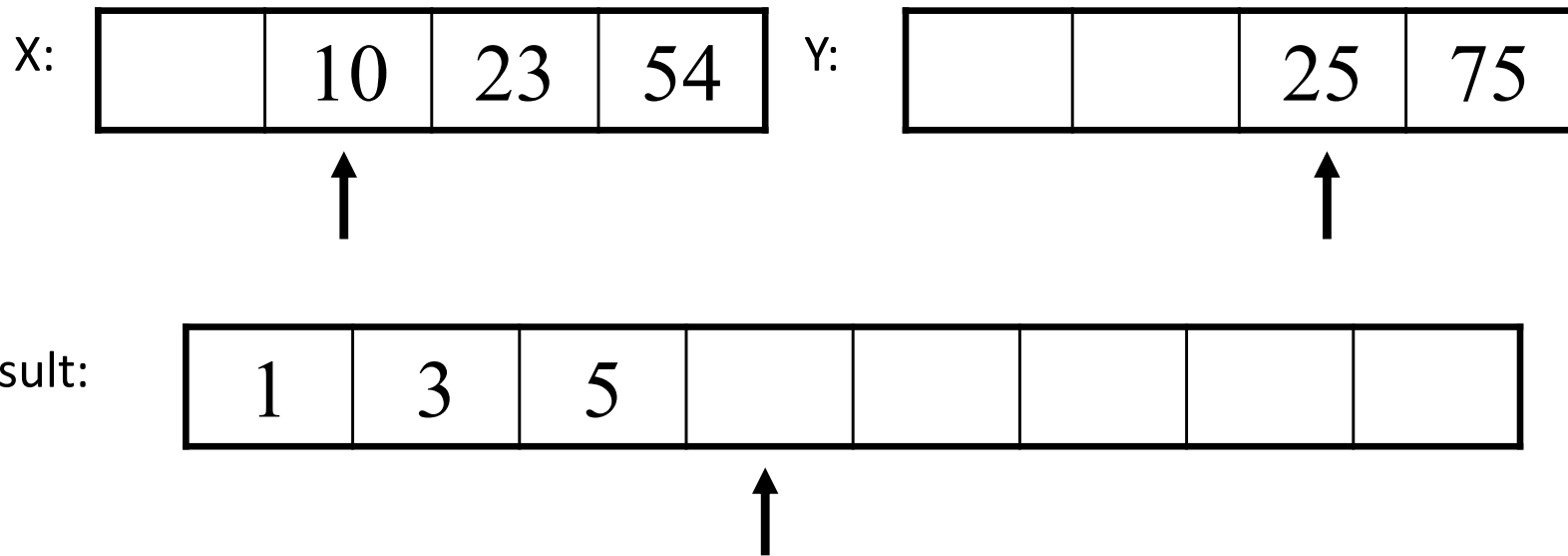

Merge Function



“No Bonus” marks for this course anymore

CPSC 204

Merge Function



“No Bonus” marks for this course anymore

CPSC 204

Merge Function

X:

		23	54
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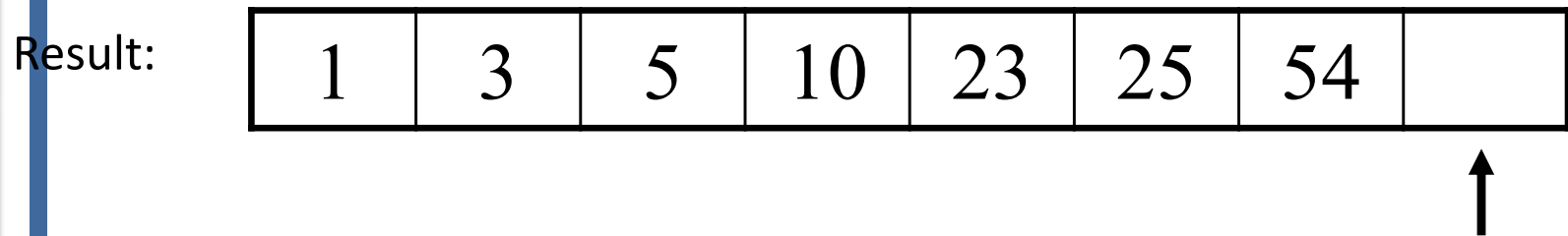
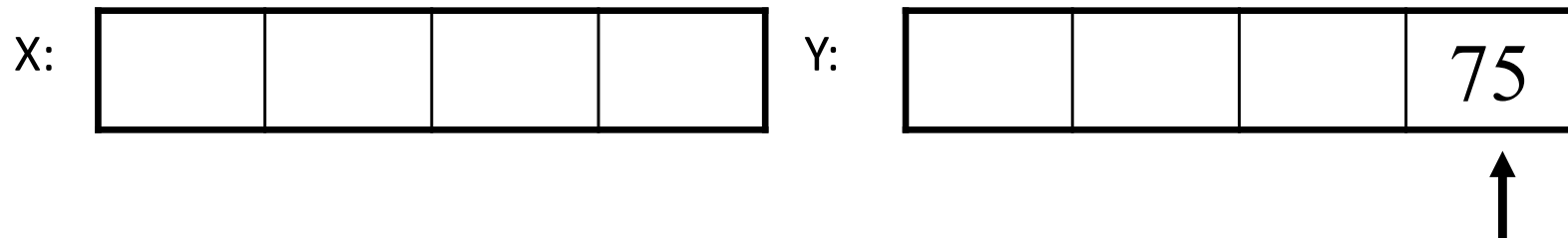
Y:

		25	75
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Result:

1	3	5	10				
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Merge Function



Merge Function

X:

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Y:

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Result:

1	3	5	10	23	25	54	75
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Merge Sort

Tracing

“No Bonus” marks for this course anymore

98	23	45	14	6	67	33	42
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C
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“No Bonus” marks for this course anymore

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98	23	45	14	6	67	33	42
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“No Bonus” marks for this course anymore

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“No Bonus” marks for this course anymore

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“No Bonus” marks for this course anymore

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Merge

“No Bonus” marks for this course anymore

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Merge

“No Bonus” marks for this course anymore

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Merge

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Merge

“No Bonus” marks for this course anymore

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Merge

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Merge

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Merge

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Merge

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Merge

“No Bonus” marks for this course anymore

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Merge

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6	33	42	67
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6	14	23	33	42	45
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“No Bonus” marks for this course anymore

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4

98	23	45	14	6	67	33	42
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98	23	45	14
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6	67	33	42
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98	23
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45	14
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6	67
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33	42
----	----

98

23

45

14

6

67

33

42

23	98
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14	45
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6	67
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33	42
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14	23	45	98
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6	33	42	67
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6	14	23	33	42	45	67
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“No Bonus” marks for this course anymore

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98	23	45	14	6	67	33	42
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98	23	45	14
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6	67	33	42
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98	23
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33	42
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23	98
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6	67
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33	42
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14	23	45	98
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6	33	42	67
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6	14	23	33	42	45	67	98
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“No Bonus” marks for this course anymore

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98	23	45	14	6	67	33	42
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98	23	45	14
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6	67	33	42
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98	23
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45	14
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6	67
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33	42
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33

42

23	98
----	----

14	45
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6	67
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33	42
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14	23	45	98
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6	33	42	67
---	----	----	----

6	14	23	33	42	45	67	98
---	----	----	----	----	----	----	----

Merge Sort

Index	0	1	2	3	4	5	6	7
Initial	25	22	34	27	28	23	16	11
Loop 1								
Loop 2								
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final	11	16	22	23	25	27	28	34

Merge Sort

Index	0	1	2	3	4	5	6	7
Initial	25	22	34	27	28	23	16	11
Loop 1	<u>22</u>	<u>25</u>	34	27	28	23	16	11
Loop 2								
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final	11	16	22	23	25	27	28	34

Merge Sort

Index	0	1	2	3	4	5	6	7
Initial	25	22	34	27	28	23	16	11
Loop 1	<u>22</u>	<u>25</u>	34	27	28	23	16	11
Loop 2	22	25	<u>27</u>	<u>34</u>	28	23	16	11
Loop 3								
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final	11	16	22	23	25	27	28	34

Merge Sort

Index	0	1	2	3	4	5	6	7
Initial	25	22	34	27	28	23	16	11
Loop 1	<u>22</u>	<u>25</u>	34	27	28	23	16	11
Loop 2	22	25	<u>27</u>	<u>34</u>	28	23	16	11
Loop 3	<u>22</u>	<u>25</u>	<u>27</u>	<u>34</u>	28	23	16	11
Loop 4								
Loop 5								
Loop 6								
Loop 7								
Final	11	16	22	23	25	27	28	34

Merge Sort

Index	0	1	2	3	4	5	6	7
Initial	25	22	34	27	28	23	16	11
Loop 1	22	25	34	27	28	23	16	11
Loop 2	22	25	27	34	28	23	16	11
Loop 3	22	25	27	34	28	23	16	11
Loop 4	22	25	27	34	<u>23</u>	<u>28</u>	16	11
Loop 5								
Loop 6								
Loop 7								
Final	11	16	22	23	25	27	28	34

Merge Sort

Index	0	1	2	3	4	5	6	7
Initial	25	22	34	27	28	23	16	11
Loop 1	22	25	34	27	28	23	16	11
Loop 2	22	25	27	34	28	23	16	11
Loop 3	22	25	27	34	28	23	16	11
Loop 4	22	25	27	34	<u>23</u>	<u>28</u>	16	11
Loop 5	22	25	27	34	23	28	<u>11</u>	<u>16</u>
Loop 6								
Loop 7								
Final	11	16	22	23	25	27	28	34

Merge Sort

Index	0	1	2	3	4	5	6	7
Initial	25	22	34	27	28	23	16	11
Loop 1	22	25	34	27	28	23	16	11
Loop 2	22	25	27	34	28	23	16	11
Loop 3	22	25	27	34	28	23	16	11
Loop 4	22	25	27	34	<u>23</u>	<u>28</u>	16	11
Loop 5	22	25	27	34	23	28	<u>11</u>	<u>16</u>
Loop 6	22	25	27	34	<u>11</u>	<u>16</u>	<u>23</u>	<u>28</u>
Loop 7								
Final	11	16	22	23	25	27	28	34

Merge Sort

Index	0	1	2	3	4	5	6	7
Initial	25	22	34	27	28	23	16	11
Loop 1	22	25	34	27	28	23	16	11
Loop 2	22	25	27	34	28	23	16	11
Loop 3	22	25	27	34	28	23	16	11
Loop 4	22	25	27	34	23	28	16	11
Loop 5	22	25	27	34	23	28	11	16
Loop 6	22	25	27	34	11	16	23	28
Loop 7	<u>11</u>	<u>16</u>	<u>22</u>	<u>23</u>	<u>25</u>	<u>27</u>	<u>28</u>	<u>34</u>
Final	11	16	22	23	25	27	28	34

Merge Sort

Analysis

C

P

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4

Merge Sort - **Algorithm**

```
method MergeSort(array) {
```

1. MergeSort(left half of array)
2. MergeSort(right half of Array)
3. Merge

```
}
```


Merge Sort - Analysis

```
method MergeSort(array) {
```

1. MergeSort(left half of array)
2. MergeSort(right half of Array)
3. Merge

```
}
```

- Let $T(n)$ be the running time of Merge Sort on an input size n
- Then we have:
 - $T(n) = (\text{Time in step 1}) + (\text{Time in step 2}) + (\text{Time in step 3})$

Merge Sort - **Analysis**

- $T(n)$: running time of Merge Sort on input size n
- Therefore, we have:
 - $T(n) = (\text{Time in step 1}) + (\text{Time in step 2}) + (\text{Time in step 3})$
- Notice that Step 1 and Step 2 are sorting problems also
 - But they are of size $n/2$...we are halving the input
- And the Merge function runs in $O(n)$ time
- Thus, we get the following equation for $T(n)$
- $T(n) = T(n/2) + T(n/2) + O(n)$
- $T(n) = 2T(n/2) + O(n)$

“No Bonus” marks for this course anymore

Merge Sort - **Analysis**

- $T(n) = 2T(n/2) + O(n)$
- For the time being, let's simplify $O(n)$ to just n
- $T(n) = 2T(n/2) + n$
- and we know that $T(1) = 1$
- So we now have a Recurrence Relation
- Is it solved?
 - NO!
- Why?
 - Damn T's!

Merge Sort - **Analysis**

- $T(n) = 2T(n/2) + n$ and $T(1) = 1$
- So we need to solve this, by removing the $T(\dots)$'s from the right hand side
- Then $T(n)$ will be in its closed form
- And we can state its Big-O running time
- We do this in steps
 - We replace n with $n/2$ on both sides of the equation
 - We plug the result back in
 - And then we do it again...till a “light goes off” and we see something

Merge Sort - Analysis

- $T(n) = 2T(n/2) + n$ and $T(1) = 1$
- Do you know what $T(n/2)$ equals
 - Does it equal 2,125 operations? We don't know!
- So we need to develop an equation for $T(n/2)$
- How?
- Take the original equation shown above
- Wherever you see an 'n', substitute with 'n/2'
- $T(n/2) = 2T(n/4) + n/2$
- So now we have an equation for $T(n/2)$

Merge Sort - Analysis

- $T(n) = 2T(n/2) + n$ and $T(1) = 1$
- $T(n/2) = 2T(n/4) + n/2$
- So now we have an equation for $T(n/2)$
 - We can take this equation and substitute it back into the original equation
- $T(n) = 2T(n/2) + n = 2[2T(n/4) + n/2] + n$
 - now simplify
- $T(n) = 4T(n/4) + 2n$
 - Same thing here: do you know what $T(n/4)$ equals?
 - No we don't! So we need to develop an eqn for $T(n/4)$

“No Bonus” marks for this course anymore

Merge Sort - Analysis

- $T(n) = 2T(n/2) + n$ and $T(1) = 1$
- $T(n/2) = 2T(n/4) + n/2$
- $T(n) = 4T(n/4) + 2n$
 - Same thing here: do you know what $T(n/4)$ equals?
 - No we don't! So we need to develop an eqn for $T(n/4)$
 - Take the eqn above and again substitute ' $n/2$ ' for ' n '
- $T(n/4) = 2T(n/8) + n/4$
- So now we have an equation for $T(n/4)$
 - We can take this equation and substitute it back the equation that we currently have in terms of $T(n/4)$

Merge Sort - Analysis

- $T(n) = 2T(n/2) + n$ and $T(1) = 1$
- $T(n/2) = 2T(n/4) + n/2$
- $T(n) = 4T(n/4) + 2n$
- $T(n/4) = 2T(n/8) + n/4$
- So now we have an equation for $T(n/4)$
 - We can take this equation and substitute it back the equation that we currently have in terms of $T(n/4)$
- $T(n) = 4T(n/4) + 2n = 4[2T(n/8) + n/4] + 2n$
 - Simplify a bit
- $T(n) = 8T(n/8) + 3n$

Merge Sort - **Analysis**

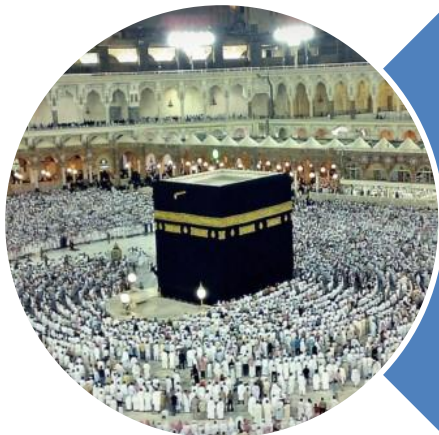
- So now we have three equations for $T(n)$:
- $T(n) = 2T(n/2) + n \quad \leftarrow 1^{\text{st}} \text{ step of recursion}$
- $T(n) = 4T(n/4) + 2n \quad \leftarrow 2^{\text{nd}} \text{ step of recursion}$
- $T(n) = 8T(n/8) + 3n \quad \leftarrow 3^{\text{rd}} \text{ step of recursion}$
- So on the k th step/stage of the recursion, we get a generalized recurrence relation:
- $T(n) = 2^k T(n/2^k) + kn \quad \leftarrow k^{\text{th}} \text{ step of recursion}$

Merge Sort - **Analysis**

- So on the kth step/stage of the recursion, we get a generalized recurrence relation:
- $T(n) = 2^k T(n/2^k) + kn$
- We need to get rid of the $T(\dots)$'s on the right side
- Remember, we know $T(1) = 1$
- So we make a substitution:
 - Let $n = 2^k$
 - and also solve for k
 - $k = \log_2 n$
- Plug these back in...

Merge Sort - Analysis

- So on the k th step/stage of the recursion, we get a generalized recurrence relation:
- $T(n) = 2^k T(n/2^k) + kn$
 - Let $n = 2^k$
 - and also solve for k
 - $k = \log_2 n$
- Plug these back in...
- $T(n) = 2^{\log_2 n} T(n/n) + (\log_2 n)n$
- $T(n) = n * T(1) + n \log n = n + n * \log n$
- So Merge Sort runs in $O(n * \log n)$ time



Allah (Alone) is Sufficient for us,
and He is the Best Disposer of
affairs (for us). **(Quran 3 : 173)**